Michael Asemota

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GitHub Project Portfolio: https://ybmikey.github.io/Michael-A-Projects/index.html
Engineering Student Seeking 2026–2027 Engineering Internships

Education

University of Texas at Arlington

Expected 2027

- Industrial Engineering Major
- Relevant Coursework: Computer-Aided Design (SolidWorks), Production & Inventory Control, Statistical Quality Control, Engineering Research Methods, Operations Research I & II, Work Measurement & Methods Engineering.

Technical Skills

- **CAD:** SolidWorks, AutoCAD, Onshape, Fusion 360, MATLAB
- **Programming:** C++, Python, Arduino
- **Manufacturing & Quality:** Lean Manufacturing, DFM/DFA, FMEA, Root Cause Analysis, Bottleneck Analysis, Fixture Design
- **Engineering Tools:** 3D Printing, Robotics, Gimbal Systems
- **Software:** Microsoft Excel, Word, PowerPoint

Certifications

- Certified SOLIDWORKS Design Associate (CSWA)
- Lean Six Sigma - White Belt

Professional Experience

Verlo (Technical Assistant)

2026 - Present

- Matched requests to optimized mattress configurations using parameters like foam density, and coil system architecture.
- Used spreadsheets to organize and analyze product data, improving collection workflows and efficiency.
- Leveraged customer and product data to identify process improvements in manufacturing and distribution efficiency.

Skechers (Associate Manager)

2022 - 2024

- Supervised daily operations for a high-volume retail store, leading 10–15 associates and increasing sales performance by 15–20% through structured coaching and performance tracking.
- Reduced stockouts by 25% by monitoring inventory and coordinating restocking using MS Office tools.
- Improved labor efficiency by 10–15% through sales data analysis and optimized scheduling, while implementing stockroom process improvements that reduced retrieval time by 30%.

Activities

NSBE (National Society of Black Engineers)

2023 - Present

- Participated in technical workshops and engineering events promoting diversity in STEM
- Engaged in collaborative engineering projects and networking opportunities with industry professionals
- Contributed to community outreach and professional development initiatives

Formula SAE

2023 - Present

- Contributed to the design, fabrication, and testing of high-performance vehicles in Formula SAE
- Applied knowledge in aerodynamics, mechanical systems, and electronics to improve vehicle performance
- Collaborated across sub-teams to troubleshoot design issues and optimize system integration

Leadership

RoboMasters (Gimbal Team Lead - 3D Modeling & Secretary)

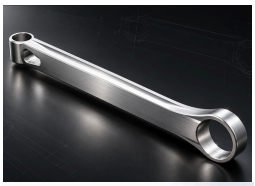
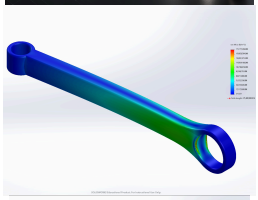
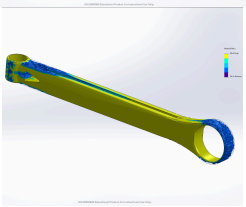
2024 - Present

- Organized and led structured team meetings, managed technical documentation, and streamlined communication workflows to support cross-functional coordination.
- Served as liaison between internal teams and external stakeholders, including sponsors and competition organizers, ensuring clear requirements alignment and project tracking.
- Directed end-to-end gimbal system development, overseeing mechanical assembly, integration, and performance optimization.
- Guided system testing and calibration efforts, improving control accuracy and operational stability through iterative validation.
- Designed and assembled a precision gimbal mechanism incorporating high-torque motors and structural components, collaborating with teammates to refine control algorithms and ensure smooth, responsive motion.

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Suspension Control Arm Design & Optimization

What

- Designed and optimized four iterations of a front lower suspension control arm for a Mercedes-Benz S550 using **SolidWorks**.
- Conducted **FEA** simulations, including static stress, fatigue, buckling, frequency, and topology studies, to evaluate structural performance.
- Refined the geometry to reduce stress concentrations, improve stiffness, and optimize material distribution.

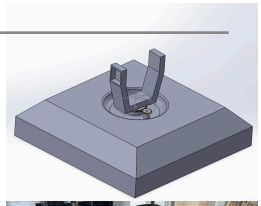
Why

- To develop a lightweight, structurally efficient suspension component while balancing strength, durability, and manufacturability.
- To evaluate how design modifications influence stress distribution, displacement, and fatigue life under simulated loading conditions.
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Results

- Reduced simulated displacement by **34.2%** while maintaining the original component mass of 1.41 lb.
- Achieved a **minimum factor of safety of 2.6**3 and an estimated **fatigue life of 15.1 million cycles** under simulated conditions.
- Completed CAD design and simulation, establishing a foundation for manufacturing feasibility analysis.

RoboMasters Standard Robot





What

- Contributed to the end-to-end mechanical design and system integration of a RoboMasters Standard Robot, including chassis structure, drivetrain architecture, and a dual-axis gimbal turret assembly.
- Collaborated cross-functionally with controls and electrical teams to integrate actuators, sensors, and targeting systems, ensuring mechanical compatibility and reliable motion performance.
- Developed and refined components using **SolidWorks** and **Fusion 360**, considering manufacturability, assembly sequence, and serviceability.

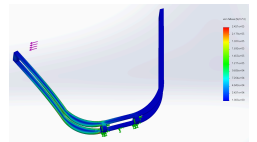
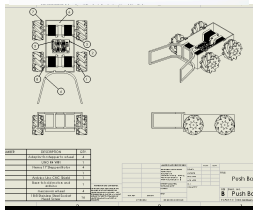
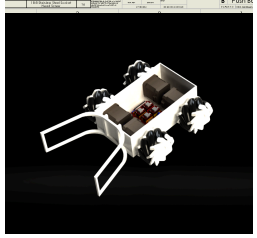
Why

- To develop a competition-ready robotic system capable of precise maneuvering, stable operation, and reliable targeting under dynamic loading conditions.
- To improve stability, accuracy, and repeatability of the turret system, enabling consistent targeting and shooting while the robot is in motion, while expanding the turret's effective range of motion.

Results

- Delivered a fully functional robot that competed in the RoboMasters 1v1 North America tournament, demonstrating stable navigation, targeting, and a scalable design suitable for future integration.

Arduino Mecanum Robot

What

- Engineered an omnidirectional robotic platform using Mecanum wheels, enabling controlled multi-axis motion and precise positioning.
- Programmed motion control algorithms in **C++ (Arduino Uno R4)** to achieve smooth acceleration, accurate direction control, and repeatable object manipulation.
- Designed the mechanical system using **SolidWorks** and **Onshape**, accounting for load distribution, wheel alignment, and drivetrain efficiency.

Why

- To design a system capable of efficient, precise motion within a constrained workspace, emphasizing controllability, repeatability, and system stability.
- To apply problem-solving and optimization techniques to overcome mechanical and control challenges.

Results

- Successfully completed the task in under 30 seconds, demonstrating high positional accuracy, reliable control, and consistent system performance.